Adaptive Intent–Preference Arbitration for Conversational Recommendation: A Reproducible Study on ReDial

Dhaya R

***Abstract*—A conversational recommender hears two voices at once: what a user has liked over many sessions, and what they are asking for in the current conversation. Most systems blend the two with a fixed rule and hope for the best. That works when the signals agree and fails quietly when they do not, for example when a habitual comedy viewer asks for a war film for the weekend. We study *Adaptive Intent–Preference Arbitration* (AIPA), a model that first classifies the relationship between the long-term preference (LTP) and the short-term intent (STI) into one of five categories (complement, consistent, conflict, override, uncertain), then chooses an action (fuse, prioritise LTP, prioritise STI, or ask a clarifying question), and finally re-ranks candidates with weights that depend on that decision. Two further components track whether an intent shift repeats across sessions and should be absorbed into the long-term profile, and probe the trained model with counterfactual ablations of each signal to report which one actually drove a recommendation. We build the whole pipeline on ReDial, an English human–human movie recommendation corpus, deriving cross-session profiles from recurring seeker identities and using MovieLens genres as item metadata. Because ReDial has no relationship annotations, labels come from a transparent weak-rule scheme and from controlled synthetic conflict injection, and every result is reported separately for the two sources. In the reduced-budget run reported here (25% of the data, 2 seeds) AIPA is clearly better than a profile-only recommender and than a GRU sequential baseline, but it does not separate from an intent-only model or from simple adaptive fusion, and the ablations are statistically flat. The arbitration head itself behaves sensibly: clarification precision is above 0.97 and wrong overrides are rare. We report these outcomes as they are, discuss why the central hypothesis is not yet supported, and list the concrete steps needed for a conclusive test. All numbers, tables and figures in this paper are generated by the released code.**

***Index Terms*—Conversational recommender systems, user intent, long-term preference, preference conflict, clarification, counterfactual analysis, reproducibility.**

I. Introduction

RECOMMENDER systems that talk to their users have an advantage that click-based systems lack: the user can simply say what they want right now. The flip side is that this fresh signal has to be reconciled with everything the system already knows about the person. Someone who has spent years watching dramas may ask, on a Friday night, for "something dumb and funny". A system that follows its profile ignores the request; a system that follows the request may throw away a decade of evidence for a single evening. Neither extreme is right, and the right answer depends on the situation.

The bulk of work on conversational recommendation (CRS) has focused on how to elicit preferences within a dialogue [1]–[4] or how to ground dialogue in knowledge about items [5], [6]. How to reconcile what is said now with what was learned before has received much less direct attention. When both signals are used, they are usually combined with a fixed rule or a learned gate that is never asked to explain itself, and evaluation almost never singles out the turns where the two disagree. Yet those are exactly the turns where the design choice matters.

This paper treats the reconciliation step as a first-class problem. We call it *arbitration*, and we make it explicit in three ways. First, the model predicts what kind of relationship holds between the long-term preference (LTP) and the short-term intent (STI): they may reinforce each other, one may add a new dimension the other lacks, they may contradict each other, the user may explicitly set the profile aside, or there may simply be too little evidence to tell. Second, this relationship drives a discrete action (fuse the signals, favour one of them, or ask a question) and the action in turn sets the fusion weights. Third, the model is made to account for its recommendations: after training we knock out one signal at a time and measure how much the ranking moves, giving a per-turn label of which signal drove the recommendation. We stress that this is a diagnostic of the fitted model, not a causal claim about users.

Our second aim is a clean and fully reproducible experimental base. We deliberately use ReDial [7], an English corpus of human–human movie dialogues. It has no cross-session profiles by design, but crowd workers recur across many dialogues, and we use that recurrence to construct chronological seeker histories under strict no-look-ahead rules. ReDial also has no labels for our relationship taxonomy; rather than pretend otherwise, we derive weak-rule labels from the dialogue text and additionally inject controlled synthetic conflicts, and we never mix the two in a table. Every claim about a metric is accompanied by bootstrap intervals and paired tests with multiple-comparison correction.

The results of the run reported here are mixed, and we say so. AIPA improves on profile-only and sequential baselines by a clear margin, its arbitration behaviour is sensible, and the counterfactual probe agrees with the chosen action on a majority of controlled cases. It does not, however, outperform an intent-only recommender or plain adaptive fusion, and none

D. R is with [affiliation]. E-mail: dhayanidhi.r@gmail.com.

Code, configuration, generated figures and the executed notebook are available at https://github.com/dhayarajas/AIPA-CRS.

of the ablations reach significance. We see the contribution of this paper as the framing, the evaluation protocol, and an honest baseline measurement that later work can build on; the discussion in Section VIII lists what a decisive test would need.

II. Related Work

A. Conversational recommendation

Early CRS work framed the dialogue as a preference elicitation game in which the system asks about attributes and the user answers [1], [2]. Estimation–Action–Reflection [3] formalised the choice between asking and recommending as a policy problem. The "system ask, user respond" paradigm of Zhang *et al.* [4] likewise treats clarifying questions as an action to be planned. Our clarification action sits in this tradition but is triggered by the relationship between two evidence sources rather than by attribute uncertainty alone.

Open-ended, natural-language CRS became tractable with ReDial [7], followed by knowledge-grounded models such as KBRD [5] and KGSF [6], and by further corpora including INSPIRED [8] and the Chinese TG-ReDial [9]. TG-ReDial is the only one of these that ships per-user histories, but its dialogues are in Chinese; we chose ReDial so that data, labels, case studies and error analysis could all be read in English, and reconstructed histories from recurring seekers instead. Surveys by Jannach *et al.* [10] and Gao *et al.* [11] give a broader map of the field; both note that reconciling the conversation with the historical profile remains open.

B. Long- and short-term signals

Session-based and sequence-aware recommenders [12]–[14] learn to weigh recent behaviour against older behaviour implicitly, through the recurrence or the attention pattern. Our sequential baseline follows this recipe. The difference in AIPA is that the weighing is exposed: the relationship class and the action are discrete, supervised (weakly) and inspectable, and the fusion weights are a function of them.

C. Counterfactual reasoning in recommendation

We borrow the vocabulary of interventions [15] for the driver diagnostic, but the object we intervene on is the trained model, not the user. Zeroing the LTP encoding and re-scoring tells us how much the model's output depends on that pathway, which is a form of ablation-based attribution. We do not claim identification of any causal effect on user behaviour, and we avoid the word "effect" when describing these numbers.

III. Problem Formulation

Consider a seeker u with a chronologically ordered set of past dialogues $\mathcal{H}_u = \{D_1, \ldots, D_m\}$ and a current dialogue D_{m+1}. At turn t of D_{m+1} the recommender introduces a new movie y_t. Our prediction target is y_t, and the evidence available is:

- **Long-term preference (LTP)**: the movies u marked as liked or seen in $D_1, \ldots, D_m$, the genre distribution over those movies, and short self-descriptive sentences ("I usually go for thrillers") extracted from earlier dialogues;
- **Short-term intent (STI)**: the seeker's utterances in D_{m+1} strictly before turn t, the movies mentioned so far in the dialogue, a genre distribution induced from those utterances and mentions, and lexical flags (negation, explicit override markers such as "for a change", explicit habit markers such as "as usual").

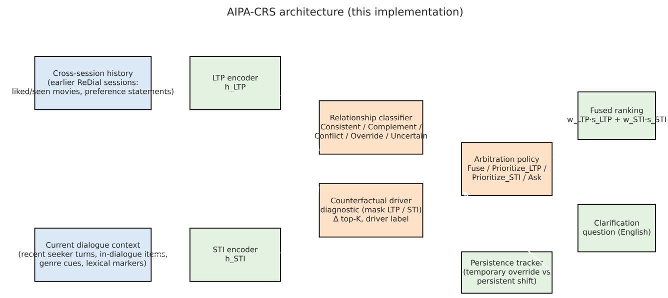

Fig. 1. AIPA components. History and dialogue context feed separate LTP and STI encoders. A relationship classifier and a counterfactual driver probe inform the arbitration policy, which produces fusion weights (or a clarifying question). A persistence tracker decides whether repeated intent shifts should update the long-term profile.

Nothing from turn t onward, and nothing from later dialogues, is visible to the model.

We define five **relationship** classes between LTP and STI. *Consistent*: STI points where LTP already points. *Complement*: STI adds a dimension that LTP does not cover but does not contradict it. *Conflict*: STI opposes LTP with no signal that the user is aware of it. *Override*: STI opposes LTP and the user explicitly flags a deliberate departure. *Uncertain*: too little evidence on either side. The **action** space is {Fuse, Prioritise_LTP, Prioritise_STI, Ask_Clarification}. A relationship implies a default action (consistent→fuse, complement→fuse, conflict→prioritise STI, override→prioritise STI, uncertain→ask), but the learned policy is free to depart from it.

We further distinguish a **temporary override**, which affects only the current session, from a **persistent shift**, in which the same departure recurs across k or more distinct sessions and should be folded into LTP.

IV. Method

Figure 1 shows the components. All encoders are small because the point of the study is the arbitration logic, not representational power; the same item tower, dimensionality and training recipe are used for every baseline so that differences are attributable to the arbitration design.

A. Item and signal encoders

Each movie is represented by a learned embedding plus a projection of its content vector (TF–IDF over English title and genre text reduced with truncated SVD to 64 dimensions, concatenated with a genre indicator from MovieLens [16]). The **LTP encoder** pools the item embeddings of the seeker's history with exponentially decaying weights (older sessions count less), concatenates the LTP genre distribution and an encoding of profile sentences, and passes the result through a two-layer MLP. The **STI encoder** does the same with the encoded recent seeker text, the current-dialogue item mentions, the STI genre distribution and the lexical flags. Both encoders output a vector of dimension 64.

B. Relationship classifier and arbitration policy

The relationship classifier takes $[h_{\text{LTP}}; h_{\text{STI}}; h_{\text{LTP}} \odot h_{\text{STI}}; |h_{\text{LTP}} - h_{\text{STI}}|]$ together with hand-crafted comparison features (cosine similarity of genre distributions, top-genre agreement, history length, cold-start flag, lexical flags) and outputs a distribution over the five relationships. The arbitration policy consumes the relationship distribution, the same comparison features and the counterfactual features described below, and outputs a distribution over the four actions. Fusion weights are computed as

$$w_{\text{LTP}} = \sum_{a} p(a)\, \beta_a, \qquad w_{\text{STI}} = 1 - w_{\text{LTP}}, \tag{1}$$

where β_a is a learned scalar per action initialised at 0.5 for Fuse and Ask, 0.8 for Prioritise_LTP and 0.2 for Prioritise_STI. The final score is $s(i) = e_i^{\top}(w_{\text{LTP}} h_{\text{LTP}} + w_{\text{STI}} h_{\text{STI}}) + b_i$. A *rule policy* variant replaces the learned action head by the default relationship→action table; it is used as an ablation.

C. Clarification

When the chosen action is Ask_Clarification and the model's confidence in that action exceeds a threshold, the system emits an English question generated from a small template set that names the top LTP genre and the top STI cue, for example *"You usually go for dramas but you mentioned horror just now—should I stick with dramas tonight or try horror?"* Offline we cannot observe the user's answer, so we evaluate clarification by its precision (was the reference label indeed uncertain or conflicting?) and by an efficiency proxy (would the target have been reachable from the top-20 without asking?).

D. Counterfactual driver diagnostic

After training, for each test turn we re-score the candidates three times: normally, with h_{LTP} zeroed, and with h_{STI} zeroed. Let δ_{LTP} and δ_{STI} be the fractions of the original top-K list that change when the respective signal is removed. The turn is labelled *Neither-driven* if both are below $\tau = 0.1$; *STI-driven* if $\delta_{\text{STI}} \geq \tau$ and $\delta_{\text{STI}} \geq 1.5\,\delta_{\text{LTP}}$; *LTP-driven* symmetrically; and *Jointly-driven* otherwise. The two δ values also enter the policy as features, so the model can learn to prefer clarification when both signals are pulling the list in different directions. Again, these are statements about the model's sensitivity, not about the user.

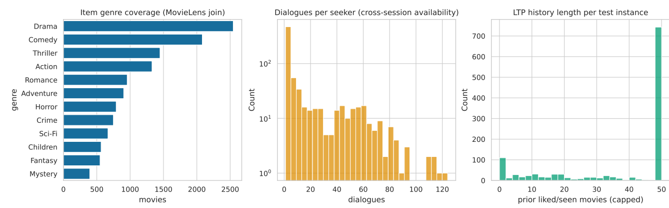

Fig. 2. ReDial overview: dialogue counts per split, dialogues per seeker, genre coverage after the MovieLens join, and turns per dialogue.

E. Temporary versus persistent shifts

The persistence tracker records, per seeker, the genre that won arbitration whenever the predicted action was Prioritise_STI. If the same genre wins in at least $k = 2$ distinct sessions, later instances of that seeker receive an LTP genre vector shifted toward it, and the event is logged. Shifts that do not recur remain session-local. In the offline setting this is evaluated by counting detected shifts and by checking that the full model and the model with the tracker disabled differ only on affected seekers.

F. Training

The recommendation loss is a sampled softmax over the item catalogue. When labels are available, a cross-entropy term on the relationship head (weight $\lambda_{\text{rel}} = 0.5$) and one on the action head ($\lambda_{\text{act}} = 0.3$) are added, each weighted per instance by the label confidence so that low-confidence weak labels contribute less. We train with AdamW [17], learning rate 0.003, batch size 128, gradient clipping at 1.0, and early stopping on validation Hit@10 with a patience of three epochs. For the "without clarification" ablation, Ask_Clarification targets are remapped to Fuse during training as well as masked at inference, so the ablation is not merely a decoding constraint.

V. Data and Labels

A. ReDial and cross-session histories

ReDial [7] contains 11,348 English dialogues (10,006 training, 1,342 test) between a seeker and a recommender, with per-movie seeker annotations for *liked*, *seen* and *suggested*. We split 1,000 training dialogues off as validation. The corpus has 764 distinct seeker identifiers and most seekers appear in many dialogues (Fig. 2). We order a seeker's dialogues by conversation identifier and treat this as chronological order; this is an assumption we could not verify against timestamps, and we flag it as such. For dialogue D_{m+1}, LTP is built only from $D_1, \ldots, D_m$; a seeker with fewer than 3 earlier sessions is treated as cold. Item metadata (genres, years) come from MovieLens *ml-latest* [16] joined on normalised title and year; movies without a match keep an empty genre vector rather than an imputed one.

Each recommendation instance is one (dialogue, turn, target movie) triple where the recommender introduces a movie not yet mentioned in the dialogue. Only the preceding 6 turns of context and at most 50 history items are used. In the run reported here we sample 25% of the dialogues in every split with a fixed seed, which yields 9281 training instances and 1077 natural test instances.

TABLE I
RELATIONSHIP LABEL COUNTS BY SOURCE AND SPLIT.

Source	Relationship	Train	Test
weak_rule	Complement	1342	165
weak_rule	Conflict	820	48
weak_rule	Consistent	4113	514
weak_rule	Override	83	14
weak_rule	Uncertain	1840	336
synthetic_controlled	Conflict	358	39
synthetic_controlled	Consistent	405	47
synthetic_controlled	Override	320	39

TABLE II
NATURAL TEST SET ($n = 1077$, MEAN OVER 2 SEEDS; BRACKETS ARE 95% BOOTSTRAP INTERVALS). ROWS ABOVE THE RULE ARE BASELINES.

Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.052 [0.044, 0.062]	0.029	0.023	0.077	0.036
STI-only	0.101 [0.089, 0.113]	0.053	0.039	0.147	0.065
Naive fusion	0.087 [0.078, 0.099]	0.048	0.036	0.132	0.059
Adaptive fusion	0.091 [0.081, 0.104]	0.051	0.039	0.134	0.062
Sequential (GRU)	0.064 [0.054, 0.074]	0.037	0.029	0.103	0.047
Conversation-aware	0.076 [0.066, 0.085]	0.045	0.036	0.122	0.057
AIPA w/o relationship	0.100 [0.089, 0.115]	0.056	0.043	0.143	0.067
AIPA w/o counterfactual	0.094 [0.084, 0.105]	0.052	0.039	0.141	0.064
AIPA w/o clarification	0.089 [0.078, 0.101]	0.049	0.037	0.126	0.058
AIPA w/o persistence	0.089 [0.078, 0.101]	0.051	0.040	0.141	0.064
AIPA (rule policy)	0.085 [0.074, 0.097]	0.046	0.035	0.126	0.057
AIPA (full)	**0.089 [0.078, 0.101]**	**0.051**	**0.040**	**0.141**	**0.064**

B. Relationship labels

ReDial has no relationship annotations. We use two label sources and keep them apart throughout.

Weak-rule labels. A deterministic procedure assigns a relationship, an action and a confidence from the LTP and STI genre distributions and from lexical evidence: explicit habit markers yield *Consistent*; negation of a habitual genre or a known tension pair (e.g. comedy vs. horror) with low genre similarity yields *Conflict*, or *Override* when an explicit departure marker is present; a new genre dimension with otherwise compatible distributions yields *Complement*; absence of genre evidence yields *Uncertain*. These labels are noisy, and the confidence attached to each is used to down-weight it in training and to define the "high-confidence" evaluation subsets.

Synthetic controlled labels. For 0.15 of natural test instances whose seeker has a usable LTP, we append one English seeker utterance requesting a genre that conflicts with, overrides, or agrees with the seeker's top habitual genre, at one of three intensities (from a mild "maybe" to an insistent request), and replace the target by a movie of the requested genre. Each synthetic instance records its source instance, the injected utterance, the intended relationship and the intensity, and carries an explicit synthetic flag. Synthetic instances appear in the training set as well (with the same flag) so that the relationship head sees clean examples of the rarer classes.

Table I gives the resulting distribution. *Override* is rare in natural data, which is expected: people seldom announce that they are departing from their habits. A slot for human-verified labels exists in the pipeline; no such annotations were available for this run and every table that would have used them reports "not run".

VI. EXPERIMENTAL SETUP

A. Compared systems

LTP-only and *STI-only* score items from one encoder. *Naive fusion* averages the two encodings with $\alpha = 0.5$; *Adaptive fusion* learns a scalar gate from the two encodings but has no relationship or action head. *Sequential (GRU)* runs a GRU [12] over the concatenation of history items and current-dialogue mentions. *Conversation-aware* attends over the current dialogue with the history vector as query. The latter two are our own re-implementations in the same parameter budget; they are not reproductions of any published system and should not be read as such. The AIPA ablations remove the relationship head, the counterfactual features, clarification, or the persistence tracker, or swap the learned policy for the rule table.

B. Metrics and statistics

Ranking quality is measured by Hit@K (equal to Recall@K with one target per instance), NDCG@K and MRR@K at $K \in \{10, 20\}$ over the full catalogue. Relationship classification is reported as accuracy, macro-F1, weighted-F1 and per-class F1; arbitration by action accuracy against the weak-rule action, conflict-resolution accuracy, override success (target in top-10 when STI is prioritised on a conflict), clarification precision, unnecessary-clarification rate and wrong-override rate. Calibration of the relationship head is measured by expected calibration error and Brier score [18]. Every metric is averaged over 2 seeds; 95% intervals are percentile bootstraps with 200 resamples over test instances; pairwise comparisons use paired t and Wilcoxon tests with Holm correction [19] within each family, and we report Cohen's d and Cliff's δ [20]. All natural-data and synthetic-data results are reported separately.

C. Compute

Everything runs on CPU (8 cores; PyTorch 2.14.0+cpu, Python 3.12.8). The configuration used here ("quick" mode) trains for at most 15 epochs with hidden size 64; the released configuration also defines a "full" mode with the complete corpus, more seeds and more bootstrap resamples, which we have not run and for which we make no claims.

VII. RESULTS

A. Overall ranking quality

Table II and Fig. 3 report the natural test set. Two things stand out. First, the intent signal dominates on ReDial: STI-only nearly doubles LTP-only, and every model that uses STI lands between 0.085 and 0.101 Hit@10. Second, AIPA (full) sits in the middle of that band. It is significantly better than LTP-only ($\Delta = +0.037$, Holm-corrected $p < 0.001$, $d = 0.18$) and than the GRU baseline on the t-test, but its differences from STI-only, naive fusion, adaptive fusion and the conversation-aware baseline are all within noise (Table III).

TABLE III

PAIRED COMPARISONS OF AIPA (FULL) AGAINST EACH OTHER SYSTEM ON HIT@10. Δ IS THE MEAN PER-INSTANCE DIFFERENCE (POSITIVE FAVOURS AIPA), p_{Holm} THE HOLM-CORRECTED PAIRED t-TEST p-VALUE (*: < 0.05), d COHEN'S d. DASHES MARK COMPARISONS WITH ZERO VARIANCE (THE PERSISTENCE ABLATION CHANGED NO TEST PREDICTION).

	Natural (all, n=1077)			Natural conflict (n=62)			Synthetic conflict (n=78)		
Control	Δ	p_{Holm}	d	Δ	p_{Holm}	d	Δ	p_{Holm}	d
LTP-only	+0.037	<0.001*	+0.18	+0.040	0.457	+0.25	+0.186	<0.001*	+0.57
STI-only	-0.012	0.251	-0.07	-0.056	0.202	-0.31	+0.000	1.000	+0.00
Naive fusion	+0.002	1.000	+0.01	-0.008	1.000	-0.07	+0.038	1.000	+0.15
Adaptive fusion	-0.003	1.000	-0.02	+0.008	1.000	+0.07	+0.019	1.000	+0.08
Sequential (GRU)	+0.025	0.001*	+0.12	+0.024	1.000	+0.11	+0.192	<0.001*	+0.59
Conversation-aware	+0.013	0.303	+0.06	-0.016	1.000	-0.10	+0.186	<0.001*	+0.56
AIPA w/o relationship	-0.012	0.303	-0.06	-0.056	0.327	-0.28	+0.026	1.000	+0.11
AIPA w/o counterfactual	-0.005	1.000	-0.03	-0.048	0.327	-0.28	-0.006	1.000	-0.02
AIPA w/o clarification	+0.000	1.000	+0.00	-0.048	0.457	-0.25	+0.006	1.000	+0.02
AIPA w/o persistence	–	–	–	–	–	–	–	–	–
AIPA (rule policy)	+0.004	1.000	+0.02	-0.016	1.000	-0.10	+0.038	1.000	+0.15

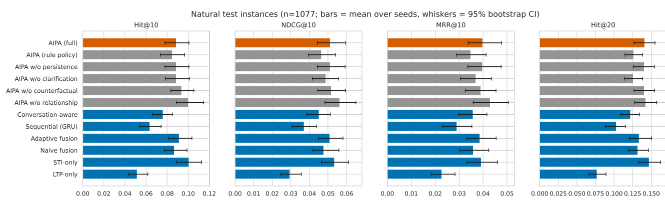

Fig. 3. Hit@10 and NDCG@10 on the natural test set with bootstrap intervals.

B. Conflict turns

The hypothesis that motivates the work is about the turns where LTP and STI disagree. Table IV shows the natural conflict subset ($n = 62$ under weak-rule labels) and the controlled synthetic conflict subset ($n = 78$), and Fig. 4 contrasts conflict with non-conflict turns.

On natural conflicts, STI-only and the relationship-free AIPA variant score highest (0.113), while AIPA (full) reaches 0.056. The confidence intervals are wide (62 instances, two seeds) and no pairwise difference survives correction, so the honest reading is "no evidence either way", with the point estimates leaning against the full model. On synthetic conflicts the picture is cleaner: every model that trusts STI does well (0.19–0.23), LTP-only and the sequential baselines collapse below 0.04, and AIPA (full) matches STI-only exactly (0.224). The synthetic set therefore confirms that AIPA learns to follow a clearly stated departure, but it also shows that "always follow the intent" is an equally good policy on this data.

Figure 5 breaks the natural set down by all five relationship classes. The *Uncertain* class (336 instances, a third of them cold seekers) is where the intent-only model gains most of its edge; *Consistent* turns, the majority, show all fused models within a few thousandths of each other.

C. Relationship classification and arbitration

Table V reports the relationship head. On natural data accuracy is 0.73–0.76 and macro-F1 0.52–0.55; the head is good at *Consistent* and *Uncertain* (F1 0.79–0.94), mediocre at *Override* (about 0.4) and poor at *Conflict* (about 0.2).

TABLE IV

CONFLICT SUBSETS, HIT@10 AND COMPANIONS. NATURAL CONFLICTS USE WEAK-RULE LABELS; SYNTHETIC CONFLICTS ARE CONTROLLED INJECTIONS. SAME LAYOUT AS TABLE II.

Natural conflict ($n = 62$)

Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.016 [0.000, 0.040]	0.013	0.012	0.032	0.017
STI-only	0.113 [0.056, 0.177]	0.046	0.026	0.210	0.069
Naive fusion	0.065 [0.024, 0.113]	0.036	0.026	0.129	0.052
Adaptive fusion	0.048 [0.016, 0.089]	0.019	0.010	0.129	0.040
Sequential (GRU)	0.032 [0.008, 0.073]	0.017	0.013	0.065	0.025
Conversation-aware	0.073 [0.024, 0.113]	0.040	0.030	0.137	0.057
AIPA w/o relationship	0.113 [0.056, 0.177]	0.052	0.033	0.161	0.064
AIPA w/o counterfactual	0.105 [0.048, 0.161]	0.052	0.036	0.161	0.066
AIPA w/o clarification	0.105 [0.056, 0.161]	0.053	0.038	0.153	0.065
AIPA w/o persistence	0.056 [0.016, 0.105]	0.033	0.026	0.153	0.058
AIPA (rule policy)	0.073 [0.032, 0.129]	0.040	0.029	0.137	0.056
AIPA (full)	**0.056 [0.016, 0.105]**	**0.033**	**0.026**	**0.153**	**0.058**

Synthetic conflict ($n = 78$)

Model	Hit@10 [95% CI]	NDCG@10	MRR@10	Hit@20	NDCG@20
LTP-only	0.038 [0.006, 0.071]	0.014	0.007	0.051	0.017
STI-only	0.224 [0.154, 0.295]	0.107	0.072	0.372	0.144
Naive fusion	0.186 [0.135, 0.244]	0.100	0.075	0.321	0.135
Adaptive fusion	0.205 [0.135, 0.269]	0.101	0.069	0.372	0.143
Sequential (GRU)	0.032 [0.006, 0.058]	0.012	0.006	0.032	0.012
Conversation-aware	0.038 [0.013, 0.071]	0.015	0.008	0.064	0.021
AIPA w/o relationship	0.199 [0.141, 0.257]	0.103	0.074	0.372	0.147
AIPA w/o counterfactual	0.231 [0.160, 0.289]	0.108	0.071	0.372	0.144
AIPA w/o clarification	0.218 [0.154, 0.282]	0.107	0.073	0.397	0.152
AIPA w/o persistence	0.224 [0.160, 0.295]	0.105	0.070	0.359	0.139
AIPA (rule policy)	0.186 [0.128, 0.244]	0.095	0.068	0.410	0.151
AIPA (full)	**0.224 [0.160, 0.295]**	**0.105**	**0.070**	**0.359**	**0.139**

Two caveats: the reference labels are themselves weak, so part of this "error" is label noise, and *Conflict* has only 48 natural test instances. On synthetic data the head is near-perfect on *Conflict* and *Override* (F1 above 0.93), which shows it can recognise the pattern when the text states it plainly; the *Complement* and *Uncertain* columns are empty there because those classes are never injected. The confusion matrix in Fig. 6 shows that natural conflicts are mostly absorbed into *Consistent*.

The arbitration head (Table VI, Fig. 7) is the most encouraging part of the results. The learned policy asks a clarifying question on about 20% of natural turns, and when it asks, the reference label is *Uncertain* or *Conflict* 98–99% of the time; unnecessary clarifications are below 0.5% and wrong overrides below 3.5%. The rule policy asks more often (27–30%) with markedly lower precision (0.69–0.75), so the learned policy

TABLE V
RELATIONSHIP CLASSIFICATION, MEAN OVER SEEDS. W-F1 IS WEIGHTED F1; THE LAST FIVE COLUMNS ARE PER-CLASS F1 (–: CLASS ABSENT FROM THE SUBSET).

Model	Subset	Acc.	Macro-F1	W-F1	Compl.	Consist.	Confl.	Overr.	Uncert.
AIPA w/o relationship	natural	0.743	0.516	0.729	0.271	0.797	0.183	0.385	0.943
AIPA w/o relationship	synthetic	0.972	0.585	0.975	–	0.974	0.953	1.000	–
AIPA w/o counterfactual	natural	0.749	0.546	0.740	0.340	0.804	0.203	0.456	0.928
AIPA w/o counterfactual	synthetic	0.960	0.582	0.971	–	0.979	0.932	1.000	–
AIPA w/o clarification	natural	0.760	0.550	0.749	0.327	0.814	0.274	0.397	0.938
AIPA w/o clarification	synthetic	0.972	0.585	0.976	–	0.980	0.953	0.994	–
AIPA w/o persistence	natural	0.727	0.524	0.725	0.301	0.785	0.194	0.410	0.930
AIPA w/o persistence	synthetic	0.972	0.587	0.979	–	0.984	0.953	1.000	–
AIPA (rule policy)	natural	0.736	0.520	0.728	0.311	0.797	0.202	0.371	0.919
AIPA (rule policy)	synthetic	0.964	0.581	0.969	–	0.969	0.938	1.000	–
AIPA (full)	natural	0.728	0.524	0.725	0.301	0.785	0.195	0.410	0.930
AIPA (full)	synthetic	0.972	0.587	0.979	–	0.984	0.953	1.000	–

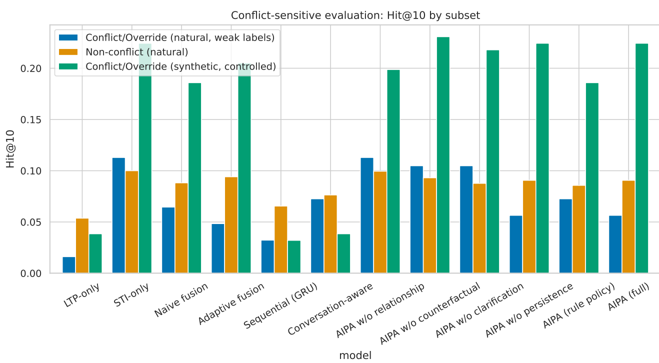

Fig. 4. Hit@10 on natural conflict versus non-conflict turns. Intervals on the conflict side are wide because the subset is small.

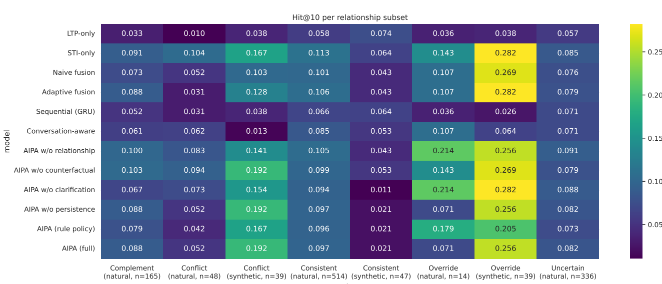

Fig. 5. Hit@10 per relationship class on natural (weak-rule) and synthetic labels.

is doing more than reproducing the table it was seeded with. Conflict-resolution accuracy on natural conflicts is low (0.15–0.26) for the reasons above; on synthetic conflicts it is 0.88–0.91. Calibration of the relationship head is acceptable (ECE 0.04–0.06 on natural data, Fig. 8).

D. Counterfactual driver diagnostic

Table VII and Fig. 9 summarise the intervention probe for AIPA (full). Removing the STI pathway disrupts the top-10 list far more than removing LTP (overlap with the original list about 0.15–0.24 versus 0.62–0.74 on natural turns), and the per-turn driver label is *STI-driven* or *Jointly-driven* almost everywhere; *LTP-driven* turns are essentially absent and no

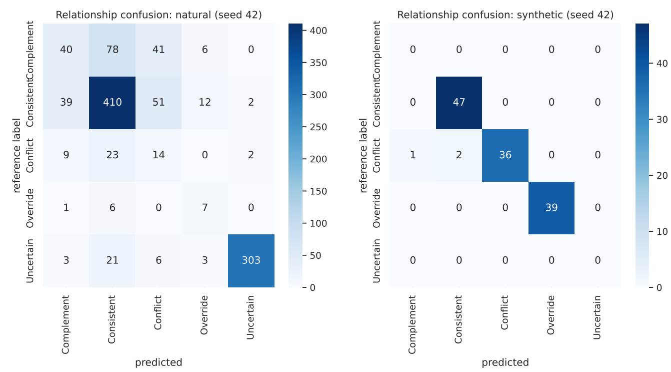

Fig. 6. Relationship confusion matrix of AIPA (full) on natural test turns (rows: weak-rule reference; columns: prediction).

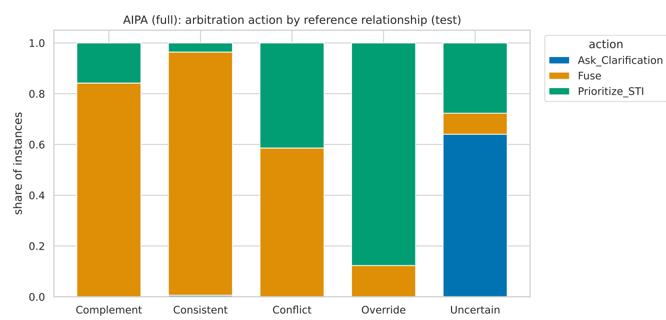

Fig. 7. Distribution of predicted actions within each reference relationship class, AIPA (full), natural turns.

turn is *Neither-driven*. On synthetic conflicts and overrides the STI dependence is strongest (overlap without STI 0.06), which is what one would want. Agreement between the driver label and the chosen action is 0.46 on natural turns and 0.62 on synthetic turns. The diagnostic thus confirms what the ranking tables already suggest: the trained model leans heavily on the conversation and uses the profile as a secondary signal.

E. Persistent shifts

The tracker flagged 33 persistent shifts across all AIPA variants and seeds (e.g. seeker 1087 repeatedly steering toward

TABLE VI
ARBITRATION AND CLARIFICATION BEHAVIOUR, MEAN OVER SEEDS. CLARIFICATION PRECISION IS UNDEFINED WHEN NO QUESTION IS ASKED.

Model	Subset	Arb. acc.	Confl. res.	Overr. succ.	Clar. rate	Clar. prec.	Unnec. clar.	Wrong overr.
AIPA w/o relationship	natural	0.881	0.161	0.383	0.204	0.987	0.003	0.012
AIPA w/o relationship	synthetic	0.932	0.891	0.256	0.000	–	0.000	0.000
AIPA w/o counterfactual	natural	0.884	0.202	0.250	0.199	0.988	0.002	0.009
AIPA w/o counterfactual	synthetic	0.932	0.891	0.269	0.000	–	0.000	0.000
AIPA w/o clarification	natural	0.688	0.153	0.583	0.000	–	0.000	0.007
AIPA w/o clarification	synthetic	0.916	0.865	0.293	0.000	–	0.000	0.000
AIPA w/o persistence	natural	0.872	0.202	0.134	0.203	0.984	0.003	0.019
AIPA w/o persistence	synthetic	0.936	0.897	0.256	0.000	–	0.000	0.000
AIPA (rule policy)	natural	0.800	0.298	0.311	0.286	0.718	0.081	0.029
AIPA (rule policy)	synthetic	0.960	0.936	0.205	0.016	0.000	0.016	0.000
AIPA (full)	natural	0.872	0.202	0.134	0.203	0.984	0.003	0.019
AIPA (full)	synthetic	0.936	0.897	0.256	0.000	–	0.000	0.000

TABLE VII
COUNTERFACTUAL PROBE OF AIPA (FULL) BY RELATIONSHIP CLASS: MEAN ABSOLUTE CHANGE IN NDCG@10 AND TOP-10 OVERLAP WHEN ONE PATHWAY IS REMOVED, AND THE SHARE OF TURNS ASSIGNED TO EACH DRIVER LABEL. THESE DESCRIBE THE MODEL, NOT THE USERS.

			$\lvert\Delta$NDCG@10$\rvert$		Top-10 overlap		Driver (%)			
Source	Relationship	n	no LTP	no STI	no LTP	no STI	STI	LTP	Joint	Neither
natural	Complement	165	0.018	0.049	0.642	0.161	50	0	50	0
natural	Conflict	48	0.020	0.030	0.642	0.142	49	1	50	0
natural	Consistent	514	0.025	0.051	0.622	0.238	46	0	53	0
natural	Override	14	0.019	0.018	0.739	0.096	54	0	46	0
natural	Uncertain	336	0.012	0.050	0.740	0.241	60	0	40	0
synthetic	Conflict	39	0.023	0.068	0.832	0.063	73	0	27	0
synthetic	Consistent	47	0.017	0.025	0.595	0.245	50	1	49	0
synthetic	Override	39	0.026	0.134	0.887	0.063	67	0	33	0

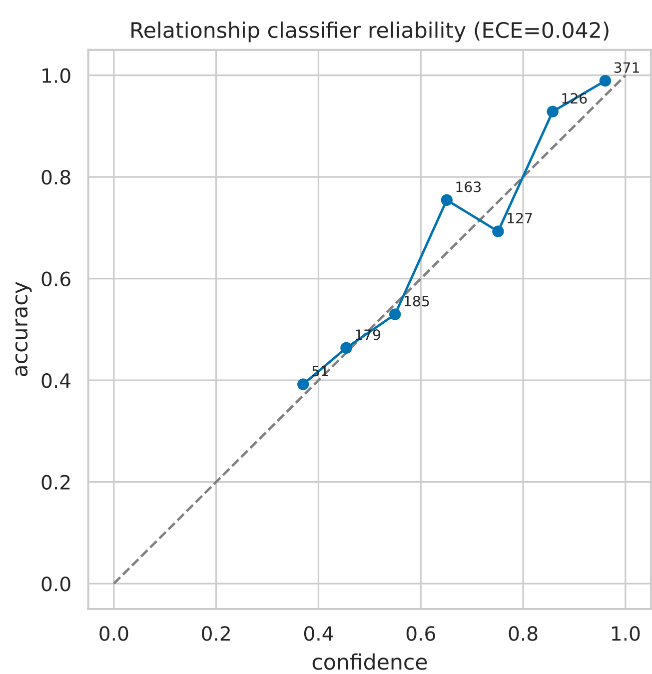

Fig. 8. Reliability diagram of the relationship head.

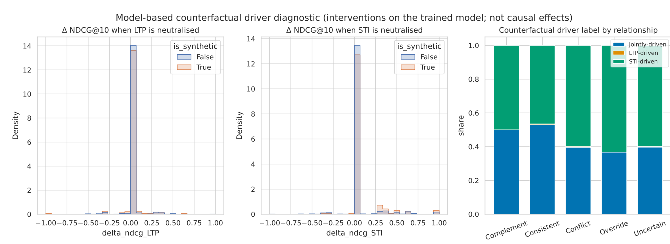

Fig. 9. Driver-label distribution and NDCG changes under signal removal.

war films, seeker 1008 toward horror). Because these shifts are only applied to instances that come *after* the second qualifying session, and few seekers reach that point inside the sampled test set, the full model and the "without persistence" ablation made identical test predictions. The mechanism works as designed but is not exercised enough by this data budget to be evaluated.

F. Sensitivity and fusion weight

Figure 10 plots Hit@10 against history length, number of seeker turns and injected conflict intensity. Performance is flat to slightly decreasing with longer histories (0.114 for 0–2 history items down to 0.085 for 26–50), a hint that long profiles add noise more than signal in the current encoder. On synthetic conflicts, AIPA (full) drops from 0.31 at the mildest intensity to 0.17 at the strongest; the ablations without counterfactual features or clarification are flatter across intensities. Table VIII shows the fixed-weight sweep for naive fusion:

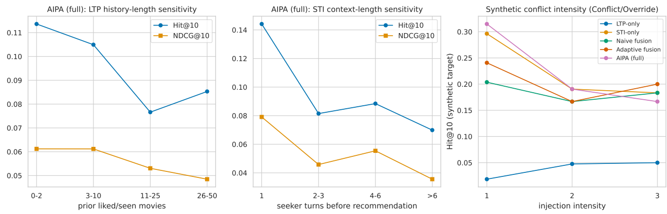

Fig. 10. Sensitivity of Hit@10 to history length, STI length and synthetic conflict intensity.

TABLE VIII
FIXED FUSION WEIGHT SWEEP (NAIVE FUSION, NATURAL TEST SET UNLESS STATED).

α_{LTP}	Hit@10	NDCG@10	Hit@20	Hit@10 (synthetic)
0.00	0.081	0.045	0.125	0.120
0.25	0.088	0.047	0.133	0.132
0.50	0.087	0.048	0.132	0.132
0.75	0.077	0.041	0.112	0.112
1.00	0.022	0.012	0.049	0.012

anything between $\alpha_{\text{LTP}} = 0.25$ and 0.5 is close to the best, and pure LTP weighting collapses to 0.022, which sets the scale for how much the profile alone can achieve here.

G. Ablations and efficiency

Reading the AIPA rows of Tables II and III as an ablation study, none of the removals changes Hit@10 by more than 0.012 and none is significant. Removing the relationship head yields the numerically best natural score (0.100), i.e. the auxiliary supervision from weak labels slightly hurts ranking in this budget. All variants have 0.48–0.55 M parameters, train in a few seconds per epoch on CPU and score a turn in under 0.05 ms (Table IX); the arbitration machinery costs roughly 6% extra parameters and a factor of two in inference time relative to naive fusion, both negligible in absolute terms.

H. Qualitative cases and error analysis

Table X shows representative test turns. The pattern is consistent with the aggregate numbers: when a seeker states a genre plainly ("war movies this weekend similar to Dunkirk"), the model reads it as an override, prioritises STI and ranks the target reasonably; when the request is indirect ("good old fashioned monster movies for my brother"), the STI genre cue is empty, the model asks, and the target sits far down the list. The most common error type is a weak-rule *Complement* or *Conflict* turn predicted as *Consistent* and fused; in the natural set 77% of *Complement* and 73% of *Conflict* references are mislabelled this way, and their targets have median ranks in the 300–550 range.

VIII. DISCUSSION

A. What the evidence supports

Three things are supported by this run. (i) Any model that reads the conversation beats a profile-only model on ReDial by a wide, significant margin; the profile alone reaches about 5% Hit@10. (ii) The learned arbitration policy is well-behaved: it asks when it should, rarely asks when it should not, and rarely overrides in the wrong direction. (iii) On plainly stated departures from habit (the synthetic set), AIPA follows the user, and the counterfactual probe shows that the ranking really is driven by the stated intent in those cases.

TABLE IX
MODEL SIZE AND COST (CPU ONLY).

Model	Parameters	Size (MB)	Train (s)	CPU inf. (ms/sample)
LTP-only	515,597	2.06	2.5	0.015
STI-only	515,597	2.06	3.1	0.026
Naive fusion	515,597	2.06	2.5	0.016
Adaptive fusion	526,862	2.11	2.8	0.022
Sequential (GRU)	492,877	1.97	8.0	0.034
Conversation-aware	476,173	1.91	1.6	0.016
AIPA w/o relationship	547,222	2.19	5.1	0.038
AIPA w/o counterfactual	547,222	2.19	5.3	0.035
AIPA w/o clarification	547,542	2.19	5.7	0.041
AIPA w/o persistence	547,542	2.19	4.0	0.039
AIPA (rule policy)	535,378	2.14	5.5	0.046
AIPA (full)	547,542	2.19	4.0	0.044

B. What it does not support

The central claim, that explicit arbitration improves recommendations over simpler fusion *specifically on conflict turns*, is not supported here. The natural conflict subset is too small for a decisive test and its point estimates lean the wrong way; on synthetic conflicts AIPA merely ties the trivial intent-following policy. The ablations are flat. We see three reasons, in rough order of importance.

First, the labels. Weak-rule relationship labels are the reference for the relationship head, the arbitration metrics and the conflict subset itself. If the rule mislabels many turns, we are training toward noise and evaluating on a subset that is not what we think it is. The near-perfect synthetic results versus the weak natural results point in that direction.

Second, the budget. A quarter of the corpus and two seeds give 62 natural conflict turns. With Hit@10 around 0.1 that is nowhere near enough to detect a plausible effect of a few points.

Third, the representation. TF–IDF/SVD text features capture genre words and little else. Much of what makes a request an intent ("for my brother", "something to fall asleep to") is invisible to the STI encoder, and so the *Uncertain* class is large and cold-start heavy.

C. What a conclusive test needs

1) Run the "full" configuration: the whole corpus, at least five seeds and 1,000 bootstrap resamples, so that the conflict subset has several hundred turns.
2) Obtain human relationship annotations for a few hundred test turns. The pipeline already accepts such a file and would report all metrics against it separately; this is the single most valuable addition.
3) Replace the text encoder by a pre-trained sentence encoder for both seeker utterances and item descriptions.
4) Add a stronger conversational baseline trained on the same inputs (a knowledge-grounded model in the spirit

TABLE X
SELECTED TEST TURNS (N: NATURAL, S: SYNTHETIC). REFERENCE LABELS ARE WEAK-RULE LABELS; "RANK" IS THE RANK OF THE TRUE TARGET UNDER AIPA (FULL).

	Dialogue excerpt (seeker turns before the target)	LTP profile	STI signal	Ref./pred. relationship	Action	$w_{\text{LTP}}/w_{\text{STI}}$	Driver	Rank
N	Recommender: Yes great decade for actions! | Seeker: Oh course! :) | Seeker: I really liked True Lies (1994) and Speed (1994) | Seeker: What do you think?	Drama 0.24; Comedy 0.17; Romance 0.16	Action 0.63; Romance 0.13; Thriller 0.13	Complement / Consistent	Fuse	0.48/0.52	STI	297
N	Recommender: Hi | Recommender: What kind of movies do you like? | Seeker: Hiya. I like rom-com movies. Can you recommend any?	Drama 0.24; Thriller 0.16; Mystery 0.11	Comedy 0.50; Romance 0.50	Consistent / Conflict	Prioritize_STI	0.33/0.67	Jointly	32
N	Seeker: Yes, I loved it | Seeker: I love movies with Will Ferrell too, like Anchorman | Recommender: What about Jumanji (2017) | Seeker: Oh, I haven't seen Jumanji (...	Crime 0.38; Drama 0.38; Film-Noir 0.09	Comedy 0.58; Adventure 0.08; Children 0.08	Conflict / Consistent	Fuse	0.45/0.55	Jointly	176
N	Recommender: hi there! I would like to reccomend some movies to ya. What kind of movies do you like? | Seeker: I;d like to see some war movies this weekend similar to ...	Comedy 0.26; Drama 0.22; Romance 0.13	War 0.68; Drama 0.18; Romance 0.06	Override / Override	Prioritize_STI	0.19/0.81	STI	2157
N	Recommender: Hello! How are you | Seeker: I'm good! My brother is coming to visit and I want to queue up some good old fashioned monster movies for him. Any suggesti...	Comedy 0.28; Horror 0.18; Sci-Fi 0.17	(no genre cue)	Uncertain / Uncertain	Ask_Clarification	0.48/0.52	STI	6182
N	Seeker: I love Blades of Glory (2007) | Recommender: Well, other good movies like that include Talladega Nights: The Ballad of Ricky Bobby (2006) | Seeker: Love Will F...	Action 0.31; Crime 0.19; Comedy 0.15	Comedy 0.75; Romance 0.25	Complement / Consistent	Fuse	0.46/0.54	STI	824
N	Seeker: American | Recommender: i could recommend you too American Pie 2 (2001) | Recommender: There are many of those movies | Seeker: Im not sure if i saw american p...	(none: cold seeker)	Comedy 1.00	Uncertain / Uncertain	Prioritize_STI	0.19/0.81	STI	726
N	Recommender: fine | Recommender: Avengers: Infinity War (2018) | Recommender: is good movie | Seeker: I saw it, it's very good	Action 0.20; Thriller 0.15; Adventure 0.14	(no genre cue)	Uncertain / Uncertain	Ask_Clarification	0.47/0.53	STI	58

of KBRD/KGSF) so that a positive result would mean something against the state of the art.

5) Tune λ_{rel}, λ_{act} and the persistence threshold; the relationship supervision currently costs a little ranking quality, and the persistence tracker was never exercised.
6) Evaluate clarification with users, or at least with simulated answers; offline precision says nothing about whether a question was welcome.

D. Threats to validity

Conversation identifiers are assumed chronological; ReDial seekers are crowd workers, not organic users, so cross-session "preferences" may partly reflect task instructions; the MovieLens join misses a fraction of titles, leaving those items without genres; and the "sequential" and "conversation-aware" baselines are our own small re-implementations. All of these are documented in the released code and reported in the generated experiment report.

IX. CONCLUSION

We set out to make the reconciliation of long-term preference and short-term intent explicit in a conversational recommender and to test whether doing so helps where it should, on turns where the two disagree. We built a complete, reproducible pipeline on an English corpus, with leak-free cross-session histories, two clearly separated label sources, a relationship classifier, a learned arbitration policy with clarification, a persistence tracker and a counterfactual probe, and we evaluated it with intervals and corrected paired tests. The reduced-budget results show a policy that behaves sensibly and a model that follows clearly stated intent, but they do not show a ranking advantage over simpler fusion, and we have not hidden that. The next step is not another model component but better labels and a larger run; the code is ready for both.

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